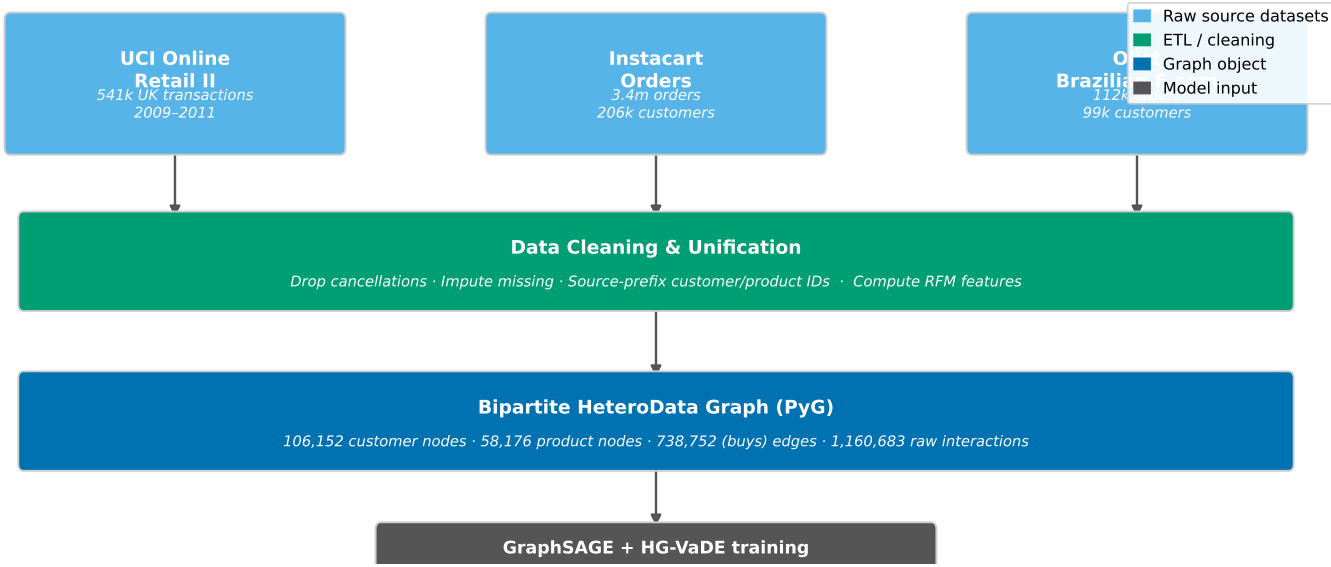
Raw source datasets
ETL / cleaning
Graph object
Model input
UCI Online Retail II
541k UK transactions
2009–2011
Instacart Orders
3.4m orders
206k customers
O
Brazilia
112k
99k customers
Data Cleaning & Unification
Drop cancellations · Impute missing · Source-prefix customer/product IDs · Compute RFM features
Bipartite HeteroData Graph (PyG)
106,152 customer nodes · 58,176 product nodes · 738,752 (buys) edges · 1,160,683 raw interactions
GraphSAGE + HG-VaDE training